

Portability and Low-Power-Drive Power Analysis

Power is not limited to the simple equation of voltage multiplied by current. From calculating the efficiency of a home appliance to determining the range of a cell tower, power measurement has a major role to play in today's systems. This article takes you through, very briefly, the latest trends in power analysers, power meters and power supplies



Power supply family from Keysight

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The growing demand for energy-efficient systems in electronics industry has led to advancements in design and test engineering. There is an increasing popularity for power measurement devices across different industry verticals, such as battery-operated portable device design, UPS and motor drive manufacturing, energy STAR power measurements, consumer appliance manufacturing and lighting. Precise and fast measurement of power and related parameters is possible with today's power supplies, power meters and power analysers. Let us explore the latest technologies and innovations in the power measurement industry that have helped our design engineers to come up with better, energy-efficient products.

Trends in the industry

While power supplies are becoming extremely efficient, the need for portable designs and low-power-consumption designs are key driving factors in power analysis segment. Given below

are the details of a few trends we find in the power measurement sector.

Faster measurement. A good measurement device responds quickly to changes in input levels, ensuring that the user gets good data even under variable conditions. Real-time measurement and analysis of power is now made possible with high-speed data delivery from the power sensors to power meters. The fast data acquisition and transfer helps in reducing the time required for measurement. This also helps make precise measurement of factors like rise time, fall time, overshoot and undershoot. Measurement speed of up to 10,000 samples per

second is achieved using models like Keysight's U2040 X-series.

High accuracy. There are applications, like mobile base-station power calculation, where one should accurately measure the power; slight variations in measured data will lead to large coverage differences. Modern power meters guarantee a high-level accuracy of approximately $\pm 0.2\text{dB}$, both in lab and field. This helps in characterising the devices more precisely, to have tighter test limits and be more accurate in fixture calibration. Power analysers like Tektronix' PA4000 offer a basic accuracy of 0.05 per cent.

Auto-ranging. Auto-ranging or auto-selecting output characteristics make a power supply much more flexible than rectangular, or traditional, output characteristic power supplies because it expands the power curve, giving the user more voltage and current combinations in a single power supply. "It is like having many rectangular power supplies in one," comments Ashish Jain, application engineer at Keysight Technologies. Power supplies such as Keysight's N8900 series and Tektronix' PA4000 series feature this capability.



MA24105A Power Sensor—Anritsu